

Uncertainty and Statistics

1 Review

The wave function tells us the probability of detecting the particle in a small detector $dP(x, t) = |\Psi(x, t)|^2 dx$

If the wave function is governed by the Schrödinger equation, the total probability is preserved. j_x is the rate that probability is flowing in the x direction at a place x at time t.

2 Uncertainty

We define an expectation value $\langle n \rangle = \sum_{n=0}^{\infty} nP(n)$, and is a constant value for a sum.

$\langle f \rangle = \sum_{n=0}^{\infty} P_n f(n)$

n is the number of "occurrences" of a thing,

Δn is the standard deviation.

Δn^2 is the variance.

$$\Delta n^2 = \langle (n - \langle n \rangle)^2 \rangle \quad (1)$$

$$= \sum_{n=0}^{\infty} P(n)(n - \langle n \rangle)^2 \quad (2)$$

$$= \sum_{n=0}^{\infty} P(n)n^2 - \sum_{n=0}^{\infty} P(n)2n\langle n \rangle + \sum_{n=0}^{\infty} P(n)\langle n \rangle^2 \quad (3)$$

$$= \langle n^2 \rangle - 2\langle n \rangle \sum_{n=0}^{\infty} P(n)n + \langle n \rangle^2 \sum_{n=0}^{\infty} P(n) \quad (4)$$

$$\Delta n^2 = \langle n^2 \rangle - 2\langle n \rangle^2 + \langle n \rangle^2 \quad (5)$$

$$\Delta n = \langle n^2 \rangle - \langle n \rangle^2 \quad (6)$$

This is how we will typically calculate the variance

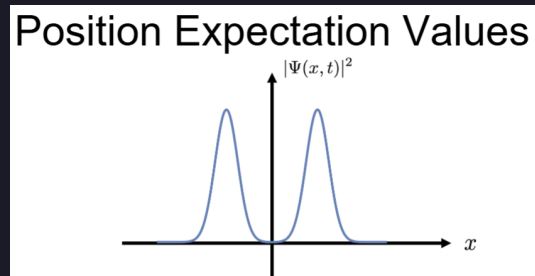
$$\Delta n^2 = \langle n^2 \rangle - \langle n \rangle^2$$

this is the weighted average of the expectation value.

We can now do this for continuous variables.

$$\langle x^a \rangle = \int_{-\infty}^{\infty} x^a dP(x) \quad (7)$$

Now lets tie in some quantum mechanics. Let's use the borne rule to calculate $\langle x^\alpha \rangle = \int_{-\infty}^{\infty} x^\alpha |\Psi|^2 dx$



Lets do a practice problem: What is the uncertainty of

$$\Psi(x, 0) = \left\{ \sqrt{\frac{30}{L^5}} x(L-x) \right\} \quad 0 < x < L \quad (8)$$

$$\begin{aligned} \langle x \rangle &= \frac{L}{2} \\ \Delta x &= ? \end{aligned}$$

$$\Delta x^2 = \langle x^2 \rangle - \langle x \rangle^2 \quad (9)$$

$$\langle x^2 \rangle = \int_0^L x^2 \frac{30}{L^5} (L-x)^2 dx = \frac{2}{7} L^2 \quad (10)$$

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} \quad (11)$$

$$= \sqrt{\frac{2}{7} L^2 - \frac{L^2}{4}} \quad (12)$$

$$\Delta x = \frac{L}{2\sqrt{7}} \quad (13)$$

We can also do this for momentum:

$$\langle P_x \rangle = \int_{-\infty}^{\infty} \frac{\Psi^* \hbar}{i} \frac{d\Psi}{dx} dx \quad (14)$$

$$\langle P_x^2 \rangle = \int_{-\infty}^{\infty} \Psi^* (-\hbar^2) \frac{d^2\Psi}{dx^2} dx \quad (15)$$

We'll derive these in big quantum but... not yet.

$$(\Delta P_x)^2 = \langle P_x^2 \rangle - \langle P_x \rangle^2 \quad (16)$$

The momentum expectation value of a Ψ that is real (doesn't have a 'meaningful' i in it. There is always some $e^{i\phi}$ phase factor, but if it is a constant or not really a function than its basically real)

$$\langle P_x \rangle = \int_{-\infty}^{\infty} \Psi \frac{\hbar}{i} \frac{d\Psi}{dx} dx \quad (17)$$

$$= \frac{1}{i} \int_{-\infty}^{\infty} \hbar \Psi \frac{d\Psi}{dx} dx \quad (18)$$

We can't have an imaginary momentum, so $\int_{-\infty}^{\infty} \hbar \Psi \frac{d\Psi}{dx} dx = 0$.

If Ψ is real, then $\langle P_x \rangle = 0$. (remember, P = momentum).

This means the distribution of momentum must be symmetric about 0.

Because of this,

$$\Delta p^2 = \langle P^2 \rangle = \langle P \rangle^2 \quad (19)$$

$$= \langle P^2 \rangle \sim \frac{\partial^2 \Psi}{\partial x^2} \quad (20)$$

This means that P depends only on the curvature of your wave function. Lower curvature has a smaller uncertainty in momentum. Therefore the more kinetic energy, the greater the uncertainty in P .

Theorem: Ehrenfest's Theorem

The correspondence principle tells us that we should expect

$$\frac{d\langle X \rangle}{dt} = \frac{p_x}{m}$$

So then the time derivative $\frac{d\langle p_x \rangle}{dt} = - \left\langle \frac{\partial V}{\partial x} \right\rangle$

Newton's second law still holds in quantum mechanics.